

A/B Testing & Experiment Analysis (Python)

Project Overview

This project demonstrates an end-to-end A/B testing framework used to evaluate the impact of a new checkout experience on user conversion and revenue. The analysis follows real-world experimentation practices, focusing on hypothesis-driven decision-making, statistical validation, and business-oriented insights.

The objective is to determine whether the treatment group (new checkout design) performs significantly better than the control group (existing checkout experience).

Business Problem

Product teams frequently introduce UI/UX changes with the goal of improving conversions and revenue. However, deploying changes without statistical validation can lead to incorrect decisions and revenue loss.

This project answers:

- Does the new checkout experience increase conversion rate?
 - Does it generate higher revenue per user?
 - Is the observed uplift statistically significant or due to randomness?
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Dataset Description

The dataset simulates real-world experiment data and includes:

- User group assignment (Control vs Treatment)
- Conversion indicator (0 = No, 1 = Yes)
- Revenue per user
- Device type
- Session duration (seconds)

The data reflects typical experimental challenges such as skewed revenue distributions and zero-inflated values.

Analysis Workflow

1. Experiment Design & Hypothesis Definition

- Defined control (A) and treatment (B) groups
- Established null and alternative hypotheses
- Set significance threshold ($\alpha = 0.05$)

2. Data Validation & Sanity Checks

- Verified balanced sample sizes
- Ensured proper randomization across devices
- Checked data integrity and experiment assumptions

3. Exploratory Data Analysis (EDA)

- Compared conversion rates visually
- Analyzed revenue per user distributions
- Examined device-level conversion behavior
- Evaluated session duration vs conversion patterns

4. Statistical Testing (Conversion Rate)

- Two-sample proportion z-test
- Assessed statistical significance of conversion uplift

5. Revenue Impact Analysis

- Revenue per user comparison
 - Welch's t-test for mean revenue
 - Mann–Whitney U test for robustness against skewed data
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Key Insights

- The treatment group achieved higher conversion rates than the control group.
- Revenue per user was higher for the treatment group.

- Results were consistent across multiple statistical testing approaches.
 - Converted users showed longer engagement durations, indicating stronger intent.
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Final Recommendation

Based on statistically significant improvements in both conversion rate and revenue per user, the new checkout experience should be rolled out to all users. Post-deployment monitoring is recommended to ensure sustained performance and detect long-term behavioral shifts.

Risk Assessment & Limitations

- Revenue data is highly skewed and sensitive to outliers.
- The experiment duration may not capture long-term effects.
- Segment-level impacts (device or geography) require further analysis.
- Results assume clean randomization and no external traffic bias.

Despite these limitations, consistent findings across tests increase confidence in the results.

Tools & Technologies

- Python
 - Pandas, NumPy
 - Matplotlib, Seaborn
 - SciPy, Statsmodels
 - Google Colab
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Deliverables

- Conversion and revenue analysis
 - Statistical hypothesis test results
 - Business-ready recommendation
 - Risk assessment for stakeholders
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Key Takeaway

This project showcases the ability to move beyond visual analysis and apply rigorous statistical reasoning to support data-driven product decisions — a critical skill for data analysts and product analysts.

Execution Environment

The entire project was developed and executed using **Google Colab**, ensuring cloud-based accessibility and reproducibility.

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